Chapter 2: Periodic Classification of Elements

Q1. State Dobereiner's law of triads and explain it with one suitable example.
Statement: Dobereiner grouped elements in triads such that when elements are arranged in increasing order of atomic mass, the atomic mass of the middle element is approximately the arithmetic mean of the other two.
Example: Li (7), Na (23), K (39)
Mean of Li and K = (7 + 39)/2 = 46/2 = 23, which is approximately the atomic mass of Na (23).

Q2. State Newlands' law of octaves. Mention any two limitations of this law.
Statement: When elements are arranged in increasing order of atomic mass, the properties of the eighth element are similar to those of the first element (like musical octaves).
Limitations (any two):
1) It was valid only up to calcium; after calcium the pattern did not match.
2) It could not accommodate newly discovered elements properly.
3) Some dissimilar elements were placed together and similar elements were separated.

Q3. Explain the main features of Mendeleev's periodic table.
Mendeleev arranged elements in increasing order of atomic mass.
Elements with similar chemical properties were placed in the same vertical column (group).
Horizontal rows were called periods.
He left gaps for undiscovered elements and predicted their properties.
Periodic table had groups and subgroups based on valency and properties.

Q4. State any four merits of Mendeleev's periodic table.
Merits (any four):
1) Systematic classification: elements with similar properties were grouped together.
2) Prediction of new elements: he left gaps and predicted properties of undiscovered elements (e.g., eka-aluminium).
3) Correction of atomic masses: some atomic masses were corrected based on proper placement.
4) Noble gases: later discovered noble gases could be placed in a separate group without disturbing the table.
5) Helped in understanding periodicity of properties.

Q5. State any four demerits (limitations) of Mendeleev's periodic table.
Demerits (any four):
1) Position of hydrogen was not fixed because it resembles alkali metals and halogens.
2) Isotopes could not be placed properly as they have same chemical properties but different atomic masses.
3) Some elements were not in correct order of atomic mass (e.g., Co and Ni) to maintain group properties.
4) No proper explanation of periodicity based on atomic structure.
5) Similar elements sometimes got separated and dissimilar placed together due to atomic mass basis.

Q6. State the modern periodic law.
Modern periodic law: The physical and chemical properties of elements are periodic functions of their atomic numbers.
That means when elements are arranged in increasing atomic number, similar properties repeat after a definite interval.

Q7. Describe the structure of the modern periodic table with reference to periods and groups.
The modern periodic table is arranged in increasing order of atomic number.
It consists of horizontal rows called periods and vertical columns called groups.
There are 7 periods and 18 groups.
Elements in the same group have similar valence shell electronic configuration and hence similar chemical properties.
Across a period, atomic number increases by 1 and new electrons are added to the same shell.

Q8. Explain how electronic configuration decides the group number of an element in the modern periodic table.
For main group elements, group number is related to the number of valence electrons.

Elements having the same number of valence electrons are placed in the same group.
Example: Elements with 1 valence electron are in Group 1 (alkali metals).
Elements with 7 valence electrons are in Group 17 (halogens).
Thus, valence shell configuration determines the group number and similar properties.

Q9. Explain the trend in atomic size (atomic radius) (a) across a period and (b) down a group.
(a) Across a period (left to right): atomic size decreases.
Reason: Nuclear charge increases, electrons are added to the same shell, so attraction increases and radius decreases.
(b) Down a group (top to bottom): atomic size increases.
Reason: New shells are added at each step, so distance of valence electrons from nucleus increases, increasing radius.

Q10. Explain the trend in metallic and non-metallic character in the modern periodic table with one example each.
Across a period (left to right): metallic character decreases and non-metallic character increases.
Reason: tendency to lose electrons decreases and tendency to gain electrons increases.
Example: In Period 3, Na is metallic while Cl is non-metallic.
Down a group (top to bottom): metallic character increases and non-metallic character decreases.
Reason: atomic size increases, so loss of electrons becomes easier.
Example: In Group 1, Li is less metallic than Cs.